INCH-POUND

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DETAIL SPECIFICATION

DISTILLER SCALE PREVENTIVE TREATMENT FORMULATIONS

Reactivated after 26 February 2007 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of general purpose distiller scale preventive treatment formulation.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

MIL-S-22698	- Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural.

FEDERAL STANDARDS

MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product.

(Copies of these documents are available online at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

T9074-BD-GIB-010/0300	- Base Materials for Critical Applications: Requirements for Low Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100.

(Copies of these documents are available from Commander, Naval Sea Systems Command, Attn.: SEA 05M2, 1333 Isaac Hull Ave. SE, Stop 5160, Washington Navy Yard, DC 20376.)

STANDARDIZATION DOCUMENT

SD-6	- Provisions Governing Qualification.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM A 213/A 213M	- Standard Specification for Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes.
ASTM A 240/A 240M	- Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.
ASTM B 167	- Standard Specification for Nickel-Chromium-Iron Alloy (UNS N06600, N06601, N06603, N06690, N06693, N06025, and N06045) and Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617) Seamless Pipe and Tube.
ASTM B 168	- Standard Specification for Nickel-Chromium-Iron Alloy (UNS N06600, N06601, N06603, N06690, N06693, N06025, and N06045) and Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617) Plate, Sheet, and Strip.
ASTM D 445	- Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity).
ASTM D 1141	- Standard Practice for the Preparation of Substitute Ocean Water.
ASTM D 1293	- Standard Test Methods for pH of Water.
ASTM D 1426	- Standard Test Methods for Ammonia Nitrogen in Water.
ASTM D 1881	- Standard Test Method for Foaming Tendencies of Engine Coolants in Glassware.
ASTM D 2688	- Standard Test Method for Corrosivity of Water in the Absence of Heat Transfer (Weight Loss Methods).
ASTM G 1	- Standard Practice for Preparing, Cleaning, and Evaluating Corrosion Test Specimens.
ASTM G 38	- Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens.

(Copies of these documents are available at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

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WATER ENVIRONMENT FEDERATION

- Standard Methods for the Examination of Water and Waste Water.

(Copies of these documents are available online at http://www.standardmethods.org/ or from the Water Environment Federation, 601 Wythe Street, Alexandria, VA 22314-1994.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The distiller scale preventive treatment formulation furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2, 4.3, and 6.3).

3.2 Material. The distiller scale preventive treatment formulation shall be in aqueous medium and shall contain organic polyelectrolytes such as polyacrylates, and an anti-foam agent that shall conform to the requirements specified herein.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.2.2 Unit content. The unit content shall be at least 6.0 gallons (or metric equivalent) of liquid at 77 °F (25 °C) as specified in 4.4.2.

3.2.3 Maintenance of heat transfer coefficient. Using the recommended dosage of 10.5 ± 0.2 milligram per liter (mg/L) the distiller scale preventive formulation in the feed water at the heat transfer coefficient shall be maintained at no less than 75 percent of the "clean tube value" throughout the 80-hour test period when tested in accordance with 4.5.1.

3.2.4 Ease of scale removal. Scale formed on heat transfer surfaces, while using the distiller scale preventive treatment formulation, shall be greater than 99.0 percent removed by one acid cleaning using the procedure given in 4.5.1.2.

3.2.5 Appearance. The distiller scale preventive treatment formulation shall be a clear liquid with no phase separation and shall form a homogeneous emulsion that is stable for 30 minutes when examined by the procedure given in 4.5.2.

3.2.6 Toxicity. The distiller scale preventive treatment formulation shall not contain toxic materials in concentrations that would constitute a health hazard. The formulation shall be safe to use for production of potable water by distillation (see 4.5.3).

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3.2.7 Corrosion. The distiller scale preventive treatment formulation shall not corrode 90-10 copper-nickel (Cu-Ni), a distiller heat transfer metal, at a rate greater than 2.54 X 10^{-2} millimeters per year (mm/yr) (1 (mil/yr)) when 90-10 Cu-Ni metal is immersed in a 100 milligram per liter (mg/L) concentration of distiller scale preventive treatment formulation in distiller water. The corrosion rate shall not be exceeded when tested in accordance with 4.5.4.

3.2.7.1 General corrosion. The distiller scale preventive treatment shall not corrode 90-10 Cu-Ni, 70-30 Cu-Ni, copper, valve bronze, phosphor bronze, gun metal or K-Monel at a rate greater than 0.10 mm/yr (4.0 mil/yr) when specimens of these metals are immersed in 100 mg/L concentration of distiller scale preventive treatment formulation in distilled water and tested in accordance with 4.5.4.1.

3.2.7.2 High temperature corrosion. Corrosion rates in excess of 0.04 mm/yr (1.5 mil/yr) for carbon steel, HY-80 steel, type 347/348 corrosion resistant steel (CRES), or Inconel Alloy 600; pitting in excess of 0.08 mm (3.0 mil) on alloy 600 and type 347/348 steel and stress corrosion cracking on stressed C-rings of Inconel Alloy 600 and 347 CRES shall not occur when tested in accordance with 4.5.4.2.

3.2.8 Compatibility with other treatment formulations. The distiller scale preventive treatment formulation shall be compatible with formulations qualified under this specification. Compatibility is the absence of a precipitate when the distiller candidate treatment formulation is mixed in the concentration recommended for use in the distiller chemical addition tank with each formulation. The testing procedure for the requirement is given in 4.5.5.

3.2.9 Homogeneity of an aqueous formulation solution. The distiller scale preventive treatment formulations shall form a stable homogeneous aqueous solution. A stable, homogeneous solution is one that does not form a phase separation after standing for 24 hours at 77 °F ± 3.6 °F (25 °C ± 2 °C) when tested in accordance with 4.5.6.

3.3 Chemical properties.

3.3.1 Ammonia content. The ammonia content of the scale preventive treatment shall not exceed 110 mg/L when analyzed in accordance with 4.6.1.

3.3.2 Hydrogen ion concentration (pH). The pH value for qualification as measured at 77 °F ± 1.8 °F (25 °C ± 1 °C) shall be not less than 6.5 or not more than 10.5 when tested in accordance with 4.6.2. The pH of subsequent production lot samples shall vary not more than 0.8 pH unit from the value obtained for the qualification sample.

3.3.3 Iron content. The iron content of the scale preventive treatment shall not exceed 10 mg/L in accordance with 4.6.3.

3.3.4 Polyelectrolyte content. The scale preventive treatment shall contain at least 14.0 percent by weight of the polyelectrolyte chemical when tested in accordance with 4.6.4. The polyelectrolyte concentration shall not vary more than 0.55 weight percent from the value obtained for qualification testing.

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3.4 Physical properties.

3.4.1 Anti-foaming action. The distiller treatment formulation shall control foaming to prevent brine carry-over into the distillate. Anti-foaming action shall meet the test requirements of 4.7.1.

3.4.2 Specific gravity. The density of the liquid shall not vary more than 0.01 grams per milliliter (g/mL) from the value obtained on qualification scale preventive treatment. Specific gravity test shall be in accordance with 4.7.2.

3.4.3 Intrinsic viscosity. The intrinsic viscosity shall be determined at 77 °F ± 0.2 °F (25.0 °C ± 0.1 °C) by the dilution method using a Cannon-Ubbelohde viscosimeter. The intrinsic viscosity shall be in accordance with 4.7.3. The intrinsic viscosity of subsequent lots shall not vary more than 0.004 liter per gram (L/g) from the value obtained when the distiller scale preventive treatment is determined for the qualification sample.

4. VERIFICATION

4.1 Classification of inspections. The inspection of the distiller scale preventive treatment formulation shall be classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests shown in table I.

4.2.1 Sample for qualification inspection. One filled 6.0 gallon (or metric equivalent) container is required for the qualification tests specified in table I.

TABLE I. Categories of inspection.

Title	Qualification	Quality conformance	Requirement paragraph	Inspection paragraph
Examination of filled containers	X	X	3.2.2	4.4.2
Maintenance of heat transfer coefficient	X		3.2.3	4.5.1 and 4.5.1.1
Ease of scale removal	X		3.2.4	4.5.1.2
Appearance	X	X	3.2.5	4.5.2
Toxicity	X		3.2.6	4.5.3
Corrosion	X		3.2.7, 3.2.7.1 and 3.2.7.2	4.5.4, 4.5.4.1 and 4.5.4.2
Compatibility with other treatment formulations	X		3.2.8	4.5.5

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TABLE I. Categories of inspection - Continued.

Title	Qualification	Quality conformance	Requirement paragraph	Inspection paragraph
Ammonia content	X	X	3.3.1	4.6.1
Hydrogen ion concentration (pH)	X	X	3.3.2	4.6.2
Iron content	X	X	3.3.3	4.6.3
Polyelectrolyte content	X	X	3.3.4	4.6.4
Anti-foaming action	X	X	3.4.1	4.7.1
Specific gravity	X	X	3.4.2	4.7.2
Intrinsic viscosity	X	X	3.4.3	4.7.3

4.3 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the QPL items in accordance with this specification and the provisions governing qualification specified in SD-6.

4.4 Conformance inspection. Conformance inspection shall consist of the applicable examinations and tests of table I.

4.4.1 Sampling for conformance inspection.

4.4.1.1 Inspection lot. For purposes of sampling, a lot shall consist of all material manufactured as one batch and transferred from one mixing tank to the shipping container.

4.4.1.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-1916 at verification level I. Verify compliance with all requirements regarding fill, closure, marking, and other requirements not requiring tests as specified in 4.4.2.

4.4.1.3 Sampling for tests. Three one-liter containers shall be filled with the distiller scale preventive formulation that shall be withdrawn from an agitated mixing tank prior to packing each lot. The samples shall be contained in plastic bottles such as polyethylene or other plastic containers that do not react with the chemicals of the distiller scale preventive treatment. The three equal portions of each sample shall be used as follows:

a. Two for conformance tests (see 4.4).
b. One for contractor.

The test results shall be submitted to the designated laboratory.

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4.4.2 Examination of filled containers. Each sample-filled container selected shall be examined for defects of the containers and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample-filled container selected shall be measured to determine the volume of the contents. To determine conformance with 3.2.2, the liquid, maintained at 77 °F ± 1.8 °F (25 °C ± 1 °C) shall be measured in a precalibrated 6.0-gallon vessel (or metric equivalent).

4.5 Test procedures for material properties.

4.5.1 Maintenance of heat-transfer coefficient. The heat-transfer coefficient, "U", shall be determined by using a calibrated model submerged tube distiller such as the 10 gallons per hour (gal/h) (38 liters per hour (L/h)) unit schematically shown on figure 1. The heat-transfer coefficient after 10 hours of operation is designated as the "clean tube value." The "clean tube value" shall be greater than 2,900 W/m^{2}k (510 BTU/f^{2}h°F). Test periods of the distiller operation shall be conducted with synthetic seawater conforming to ASTM D 1141. The distiller scale preventive treatment shall be added to the seawater in a concentration of 10.5 ± 0.2 mg/L. An additional 1 mg/L of polyethylene glycol monooleate (PEG-200-MO) anti-foam agent shall be added to the distiller feed water to control excessive brine carryover in the test unit. The temperature data needed to calculate the heat transfer coefficient shall be recorded every hour during periods of operation. The distiller shall be operated at 185 °F ± 1.8 °F (85 °C ± 1 °C) boiling temperature, and the average feed water flow rate shall be 0.5 gallon per minute (gal/min) (1.9 liters per minute. The steam supplied by the boiler shall be saturated and at a temperature ranging between 212 °F to 230 °F (100 °C to 110 °C). The heat supplied by the boiler shall be constant at a rate of 30,000 W (102,500 BTU/h). The brine concentration ratio shall be maintained at 1.5 ± 0.1. The test shall be conducted for 90 hours including the initial 10-hour break-in period.

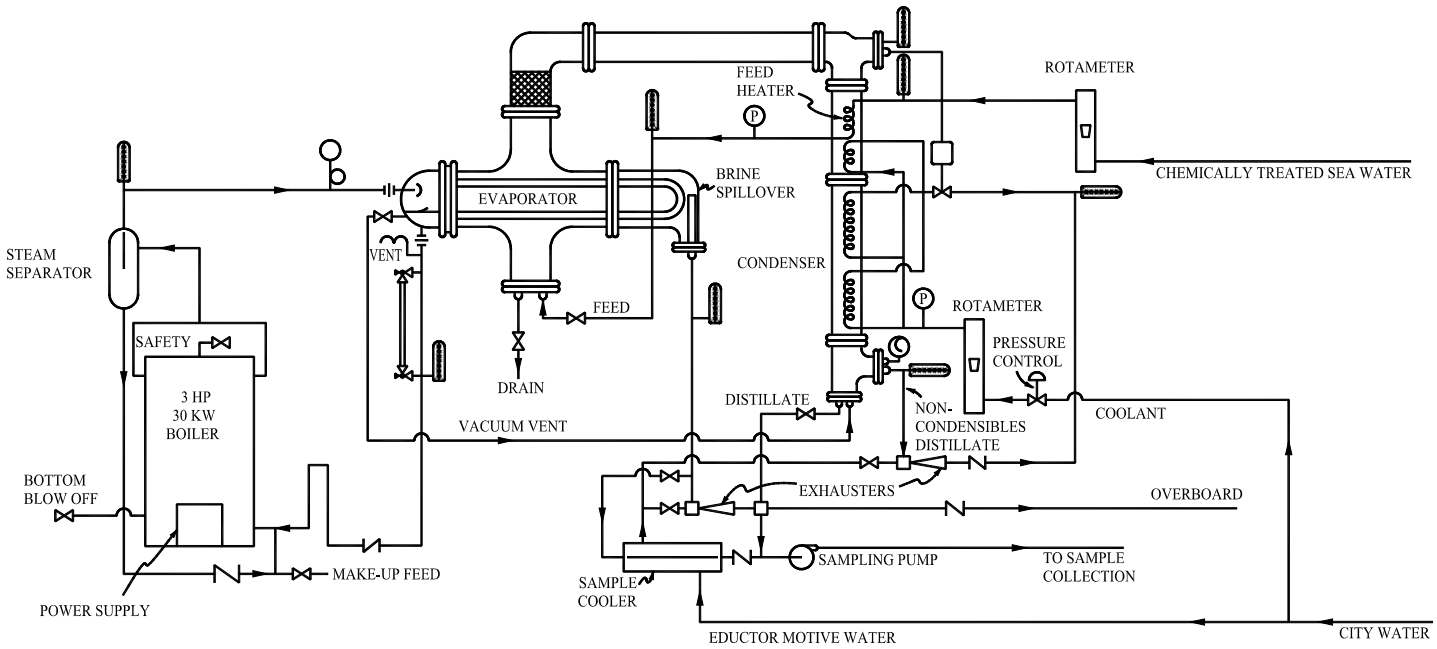


FIGURE 1. Diagrammatic arrangement-laboratory 38 L/h (10 gal/h) distillation system.

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4.5.1.1 Percent heat-transfer retention calculations. The heat-transfer coefficient, "U", shall be expressed by the following equation:

$$U = Q/A\Delta T$$

Where Q = The heat flow rate in watts (W) or BTU/h.
A = The heat-transfer surface area in units of square meters (m^2) or square feet (ft^2).
ΔT = The apparent temperature difference between the saturated steam temperature and the boiling temperature in the evaporator chamber in °C or °F.

"Q" is determined from the electrical input (30,000 W) into the boiler. The heat-transfer coefficient, "U", is affected by scaling of heat-transfer surfaces during operation of the distiller. After the first 10 hours of the test run, the heat-transfer coefficient shall be calculated by using the value of "A" for the distiller, the measured value ΔT and the "Q" value for the heat flow. The heat-transfer coefficient shall be calculated for every hour of operation. The percent heat-transfer retention shall be calculated as follows:

Percent heat-transfer retention = U_2/U_1 X 100

Where U1 = Clean tube heat-transfer coefficient W/m^2K determined after the first 10 hours of operation.
U2 = Heat-transfer coefficient W/m^2K determined at 1-hour intervals during the last 80 hours of operation.

4.5.1.2 Ease of scale removal. At the completion of the 90 hour test period, the ease of scale removal shall be determined by cleaning the distiller with 1 kilogram (kg) of inhibited sulfamic acid dissolved in 20 kg of water, making a 5 percent weight solution. This solution shall be injected into the distiller and allowed to stand for one hour; the distiller shall be heated to 200 °F (93 °C) for another hour, and then rinsed with fresh water. A visual examination of the tubes shall be made to determine the effectiveness of the acid cleaning to descale the heat transfer surfaces. No more than 1.0 percent of the 0.58 meters squared (m^2) (6.25 feet squared (ft^2)) heat transfer surface area shall have scale remaining after the acid cleaning.

4.5.2 Appearance. The distiller scale preventive treatment formulation shall be visually examined. It shall be a clear liquid or it shall be a homogeneous, stable emulsion for 30 minutes after vigorously shaking a 125-mL sample for 1 minute in a stoppered 250-mL Erlenmeyer flask.

4.5.3 Toxicity. The distiller scale preventive treatment formulation shall be certified by Bureau of Medicine and Surgery (BUMED) to be safe for use to produce potable water by seawater distillation.

4.5.4 Corrosion. A 90-10 Cu-Ni coupon shall be placed in a one-liter Erlenmeyer flask and 500 mL of distilled water containing 100 mg/L of distilled scale preventive treatment formulation shall be added. A water-cooled condenser shall be attached to the flask. The solution is heated on an electric hot plate and allowed to reflux for 30 hours. The coupon preparation and calculation of corrosion rate shall be conducted according to procedure B of ASTM D 2688.

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4.5.4.1 General corrosion. Duplicate corrosion test coupons of 90-10 Cu-Ni, 70-30 Cu-Ni, Cu, valve bronze, phosphor bronze, gun metal and K-Monel shall be placed in separate standard taper one-liter Erlenmeyer flasks. The duplicate coupons shall be separated by Teflon®, or equivalent, washers. The coupons shall be submerged in 650 mL of solution in distilled water containing 100 mg/L of distiller scale preventive treatment formulation. A water-cooled, standard taper condenser shall be attached to the flask. The solution shall be heated and maintained at 210.2 °F ± 2 °F (99 °C ± 1 °C) for 96 hours. The coupon preparation and calculation of corrosion rate shall be conducted in accordance with procedure B of ASTM D 2688.

4.5.4.2 High temperature corrosion test. The mill annealed specimens shall be tested for 4 hours per inch of thickness (4 hours minimum).

4.5.4.2.1 Mass change specimens. Two mass change coupons 1 inch X 2 inch X 1/8 inch (25 mm X 51 mm X 3.2 mm) of 347 CRES, Inconel Alloy 600, HY-80, and carbon steel shall be polished with 600 grit or finer metallurgical polishing paper, cleaned in alcohol or electronic grade acetone, dried in a desiccator, and weighed to the nearest 0.1 mg. The plate specimens shall meet requirements of MIL-S-22698 for carbon steel, T9074-BD-GIB-010/300 for HY-80, ASTM A 240/A 240M for Inconel Alloy 600, and ASTM B 168 for CRES 347/348.

4.5.4.2.2 Stress corrosion cracking specimens. Stressed C-rings (2 each) of 347 CRES and Inconel Alloy 600 shall be fabricated from 0.5 inch OD X 0.053 inch (minimum) (1.25 centimeter (cm) outside diameter (OD) X 0.135 cm) wall thickness, seamless tubing in accordance with ASTM G 38. The tubing for the C-ring specimens shall meet requirements of ASTM B 167 for Inconel Alloy 600 and requirements of ASTM A 213/A 213M for the type 347/348 CRES. The specimens shall be 0.5 inch (1.27 cm) OD with a minimum wall thickness of 0.053 inch (0.135 cm). Tubing specimens shall be fabricated from mill annealed tubing material that has been stress relieved at 1,125 °F ± 25 °F (607 °C ± 14 °C) for 4 hours. The specimens shall be loaded beyond the proportional limit to have a maximum plastic strain of 5 percent to 10 percent. Stressed specimens shall receive no subsequent heat treatment.

4.5.4.2.3 Pre-test examination. Fabricated specimens should receive a pre-test examination to ensure that no unacceptable pitting or cracking exists on the specimens prior to testing.

4.5.4.2.4 High temperature corrosion test procedure. The mass change coupons shall be mounted in Teflon®, or equivalent, holders and these, along with the stressed C-rings, shall be suspended in a CRES static autoclave vessel. The latter shall be filled with enough aqueous solution containing 12,500 mg/L (p/m) distiller scale preventive treatment to assure at least a 25 mm layer of solution above all the specimens and to allow for expansion due to heating. The solution of distiller scale preventive treatment shall be analyzed and reported for pH, conductivity, chloride, and sodium before and after the test to determine if the solution changed during the testing. The sealed autoclave shall be heated to 584.6 °F ± 9 °F (307 °C ± 5 °C) and maintained at this temperature for a continuous 720-hour period. After removing the specimens from the autoclave, each coupon shall be chemically cleaned and the corrosion rate of each mass change coupon shall be measured in accordance with ASTM G 1. Following the final chemical cleaning, each mass change coupon shall be rinsed in accordance with ASTM G 1, with either alcohol or

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electronic grade acetone, dried in a desiccator, and then weighed to the nearest 0.1 mg. Corrosion rates shall be calculated in accordance with ASTM G 1. C-rings shall be examined using a microscope at low magnification (about 50X) for stress corrosion cracks. The entire surface of each specimen shall be examined. In addition, all C-rings shall be sectioned transversely, mounted, and metallographically examined for cracks at magnification up to 500X. No stress corrosion cracking shall be evidenced as determined by surface and metallographic examination.

4.5.5 Compatibility with other treatment formulations. One hundred mL of a 10 percent solution of the distiller scale preventive treatment formulation shall be mixed at 77 °F ± 1.8 °F (25 °C ± 1 °C) with 100 mL of a 10 percent solution of each formulation previously qualified. The material being tested shall not be considered compatible if a precipitate forms when the two solutions are mixed.

4.5.6 Homogeneity of an aqueous formulation solution. Prepare a solution of the distiller scale preventive formulation by adding 7.5 mL of a well-mixed sample to a 500-mL volumetric flask. Add distilled water to the mark. Shake vigorously for 1 minute to homogenize the solution. Allow the solution to stand for 24 hours at 77 °F ± 3.6°F (25 °C ± 2 °C). No phase separation shall occur.

4.6 Test procedures for chemical properties.

4.6.1 Ammonia content. No greater than 110 mg/L of ammonia shall be present in a sample of the liquid. The analysis shall be made according to ASTM D 1426 by the distillation method using the Nessler reagent.

4.6.2 pH value. The pH of the liquid shall be greater than 6.5 and less than 10.5 when measured using the method described in ASTM D 1293. The pH shall be within 0.8 pH unit of the value obtained as specified in 3.3.2.

4.6.3 Iron content. The iron concentration of a sample shall be determined by the phenanthroline method described in part 3500-Fe Iron of "Standard Methods for the Examination of Water and Waste Water" and shall not exceed the iron content limit given in 3.4.3.

4.6.4 Polyelectrolyte content (polymer content - acetone-insoluble material). A 2 mL portion of the well-mixed liquid formulation at 77 °F ± 1.8 °F (25 °C ± 1 °C) shall be measured by a volumetric pipette and transferred to a dried, preweighed 100 mL centrifuge tube plus a stirring rod. The centrifuge tube shall be weighed with an accuracy of ±0.2 mg. Twenty-five mL of reagent grade acetone (water content of 0.5 percent maximum) shall be added to the sample and the mixture stirred with the stirring rod for 2 or 3 minutes. The stirring rod shall be removed from the tube and placed across the top of a beaker, being careful not to loosen any of the adherent material. The tube shall be capped and centrifuged for 5 minutes at a speed at which the centrifuge is capable of producing a relative centrifugal force (ref) between 600 and 700 at the tips of the tubes. The supernatant liquid layer shall be removed from the polymer residue. A second 25-mL portion of acetone shall be added to the residue and the extraction procedure repeated. The acetone extraction shall be repeated with a third 25-mL portion of acetone. The solid polymer material remaining in the 100-mL centrifuge tube shall be gently warmed on a hot plate (surface temperature 140 °F (60 °C)) in a well-ventilated hood to remove the bulk of the

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remaining acetone. The stirring rod shall be inserted into the tube, and the solid shall be dried in an oven at 221 °F ± 3.6 °F (105 °C ± 2 °C) for 1 hour. The solid shall then be cooled in a desiccator over a suitable desiccant and the tube and stirring rod shall be weighed. The drying shall be repeated until successive weighings are within ±1 mg. The percentage of acetone insoluble polymer shall be within ±0.55 mass percent of the value obtained as specified in 3.3.4. The percentage of acetone insoluble shall be calculated as follows:

$$\text{Percent acetone insoluble material} = \frac{\text{Mass of residue in centrifuge tube}}{\text{Mass of original liquid sample}} \times 100$$

4.7 Test procedures for physical properties.

4.7.1 Anti-foaming action. The test is a modification of ASTM D 1881. A gas dispersion tube is used that is 250 mm long and has a 12 mm outside diameter coarse fritted glass cylinder. A flow meter is placed between a tank of breathing-quality compressed air and the bubbler. A flow of 1 L/min is regulated by the pressure valve on the tank. A vigorously shaken 250-mL sample of liquid formulation and 1 molar (M) sodium chloride solution are used to prepare a 1 percent solution that is 0.2 M with respect to sodium chloride. The test shall be performed at a temperature of 194 °F to 203 °F (90 °C to 95 °C). With these modifications, the ASTM procedure is then followed. No foam should persist immediately after the air is secured; that is, the "eye" should form immediately.

4.7.2 Specific gravity. A clean, dry 25-mL pycnometer shall be weighed. Mass shall be recorded to the nearest 0.1 mg. The stopper shall be removed and the pycnometer filled with recently boiled distilled water that has been cooled. The stopper shall be inserted, any excess water removed, and the pycnometer allowed to reach 77 °F (25 °C). The pycnometer shall be weighed and the weight of the water recorded to the nearest 0.1 mg. The procedure shall be repeated by using the well-mixed liquid distiller treatment formulation. The specific gravity of the distiller treatment shall be calculated as follows:

$$\text{Specific gravity} = \frac{\text{Mass of formulation - mass of pycnometer}}{\text{Mass of water - mass of pycnometer}}$$

The specific gravity shall meet the requirement specified in 3.4.2.

4.7.3 Intrinsic viscosity. The intrinsic viscosity shall be determined as follows:

a. The intrinsic viscosity shall be determined at 77 °F ± 0.2 °F (25 °C ± 0.1 °C) by the dilution method using a Cannon-Ubbelohde four-bulb shear dilution viscometer with a shear range at the wall of 45 to 1,800 sec^{-1} or equivalent. The first step shall be to determine the reduced specific viscosity. For the necessary calculations of the reduced specific viscosity, a concentration of 100 grams per liter (g/L) for the active ingredient in the scale preventive formulation shall be assumed. Using a well-mixed sample, three dilutions shall be prepared by pipetting 10mL for a 20 g/L solution, 7.5 mL for a 15 g/L solution, and 5 mL for a 10 g/L solution into separate 50-mL volumetric flasks. Add 10 mL of 1M sodium chloride solution to each flask and bring to the mark with distilled water to produce a final concentration of 0.2 M sodium chloride. A fourth flask containing 0.2 M sodium chloride and no scale preventive formulation shall be prepared.

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b. Flow-times for the three dilutions (starting with the least diluted solution) and 0.2 M sodium chloride shall be determined after proper precautions are taken to clean the viscometer in accordance with ASTM D 445. The reduced specific viscosity, expressed in L/g, shall be calculated as follows:

$$\eta_{sp} = \frac{t_c/t_o - 1}{C}$$

Where η_{sp} = The reduced specific viscosity.
t_c = The flow through times obtained with the three diluted samples.
t_o = The time obtained with the 0.2 M sodium chloride solution.
C = The concentration of the sample active ingredients in g/L.

The shear rates at the wall ($\dot{\gamma}$) expressed in reciprocal (sec^{-1}) shall be determined as follows:

$$\dot{\gamma} = \frac{k}{t_c}$$

Where k = Calibration constant of the viscometer.
t_c = The flow time.

A plot of η_{sp} versus $\dot{\gamma}$ (four points) shall be constructed for each diluted sample, and the reduced specific viscosity is determined at a shear rate of 1,000 sec^{-1} or where the curve is closest to being horizontal. The equation for the best straight line for the points representing the reduced specific viscosities for the three concentrations (10, 15, and 20 g/L) shall be calculated using the least square method and the intrinsic viscosity shall be the y-intercept of this line. The intrinsic viscosity shall meet the requirements of 3.4.3.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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5.2 Marking data.

a. Manufacturer's or contractor's name or both.
b. Lot number.
c. Contract or order number.
d. Date of manufacture (month and year).
e. Precautionary labeling.
f. Shelf life (expiration date of 48 months from date of manufacture).
g. "MANDATORY SHELF LIFE MATERIAL, DISCARD MATERIAL AFTER EXPIRATION DATE".
h. Do not store in direct sunlight.
i. For transfer at sea operations, other than full pallet loads, breakout units should be places on the top of the transfer load.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This distiller scale preventive treatment chemical covered by this specification is for scale prevention for submerged tube, basket, and flash-type naval distillers. The distillate for distillers using this treatment is used for potable water and makeup feed water for all Navy boilers. A dosage rate of one pint of scale preventive treatment per 4,000 gallons of distillate is standard for Navy shipboard use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. Packaging requirements (see 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-24577 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Richmond, Attn: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297 or STDZNMGT@dla.mil.

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6.4 Subject term (key word) listing.

anti-foam
boilers
distillate
makeup feed
polyacrylates
potable
seawater
water

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy - SH

Preparing Activity:
DLA - GS3

(Project 6850-2005-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil.

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